Zhiying Tang

Phone: +86 18302044150 | Email: Zhiying.Tang23@student.xjtlu.edu.cn | GitHub: github.com/Katherine-Tang

EDUCATION

Xi'an Jiaotong-Liverpool University (XJTLU) Sept 2023 – Present
Bachelor of Science in Information and Computer Science
Dual Degree Programme with the University of Liverpool (UK)

- **Coursework:** Java Programming, Introduction to Databases, Software Engineering
- **GPA:** 3.92/4.0

CAMPUS EXPERIENCE

Student Representative Union, Vice President Sept 2023–Present

- Led student engagement initiatives across multiple colleges, organizing **workshops and cross-school events** with 100+ participants.
- Chaired the **Student Representative Conference**, coordinating communication between student representatives and university administration to address academic and campus-life concerns.
- Led the collection and consolidation of **50+ student proposals**, successfully advocating for the implementation of **12 concrete improvements** by the university.
- Independently planned and moderated **3 cross-college faculty–student forums**, producing **15 actionable recommendations** for course and curriculum enhancement.
- Collaborated with department leaders to **collect student feedback**, prepare reports, and improve transparency in school–student communication.

RESEARCH EXPERIENCE

Summer Undergraduate Research Fellowship — FX Risk Sentinel Analysis June 2025 – Sept 2025
Supervisor: Dr. Ren Zhao (School of Mathematics and Physics) XJTLU

Research Overview
Participated in the development of an FX risk prediction system designed to forecast foreign exchange risk spikes and USD/CNH volatility. Contributed to the full 0–1 development pipeline—from data collection and preprocessing to the end-to-end deployment of both the iOS application and web platform. Enhanced predictive performance through model architecture refinement, cross-frequency feature alignment, and the integration of mixed-frequency econometric indicators.

Technical Highlights

- Collected and merged multi-source financial datasets from **Wind, Bloomberg APIs**, and public economic database.
- Applied the **MIDAS** algorithm to integrate mixed-frequency data, aligning daily FX indicators with monthly and quarterly macroeconomic factors for improved temporal consistency.
- Built and optimized **XGBoost** and **Random Forest** classification models for Forex risk level prediction, achieving 75% precision with feature importance analysis (volatility, inflation signals, capital flows).
- Performed model evaluation using RMSE, MAE, MAPE, and R^2 metrics, demonstrating improvements in medium/long-horizon forecasting.
- Built a production-ready **iOS** mobile application (SwiftUI) enabling real-time FX risk scoring with API-based model inference and live market data retrieval. Achieving version switch on English and Chinese.
- Designed and managed **MySQL** databases for structured storage of macroeconomic datasets, model outputs, and user interaction logs; developed a responsive HTML dashboard for risk visualization and data querying.

Key Achievements

- Helped the team achieve strong model performance: RMSE = 5.41×10^{-5}, and R^2 = 90.23%.
- Contributed to the system's validated ability to detect FX risk spikes across March 2024–March 2025, particularly surrounding tariff tensions and geopolitical shocks.

- Delivered two end-to-end products: a **web dashboard** and a fully functional **iOS app** supporting multilingual mode and real-time data streaming.

Tech Stack

- Python, XGBoost, Random Forest, MIDAS Regression, FinBERT, HTML, MySQL, Wind Terminal, LaTeX
- iOS Development: SwiftUI, Xcode, RESTful APIs(ExchangeRate API, DeepSeek API)

PROJECTS EXPERIENCE

Contemporary Undergraduate Mathematical Contest in Modeling — Sept 2024

Technical Writing & Modeling Documentation Lead

Project Overview

Designed and implemented a multi-objective crop planting optimization system for rural agricultural planning using real-world farmland and crop datasets. The project integrates mathematical modeling, bio-inspired algorithms, and stochastic simulation to support crop allocation and sales strategy optimization under multiple constraints and uncertain conditions.

Technical Highlights

- Processed and integrated multi-source agricultural data including land plots, crop attributes, yield, cost, and price using Python (Pandas, NumPy) and Excel for visualization and statistical analysis.
- Developed a multi-objective **grey wolf optimizer (MOGWO) algorithm** to maximize profit and minimize planting dispersion under real agricultural constraints such as crop rotation, replanting restrictions, and seasonal planting rules.
- Incorporated **Monte Carlo model** simulation to model uncertainties in crop yield, market price, planting cost, and climate impacts for robust decision-making.
- Analyzed crop substitutability and complementarity, and applied **Pearson correlation** to explore relationships among sales volume, price, and cost.
- Wrote the final modeling paper using **LaTeX**, ensuring professional typesetting of mathematical formulations, algorithms, and result tables.

Key Achievements

- Delivered robust financial projections by modeling two distinct sales scenarios, successfully estimated the total profits at the case.
- Engineered a dynamic planning solution that automated the output of optimal seasonal planting arrangements across all 54 land plots over 7 years, leveraging a Python-based optimization algorithm.
- Communicated complex analysis with clarity by authoring a comprehensive, professionally typeset LaTeX report, effectively presenting the model's methodology, constraints, and strategic insights.

Tech Stack

- Python, Excel, MOGWO (Multi-Objective Grey Wolf Optimizer), Monte Carlo Simulation, LaTeX

IEEE CyberC Big Data Competition — Oct 2024

Data Analysis & Feature Engineering — HKUST (Guangzhou)

Project Overview

Designed and implemented a city traffic flow forecasting and intelligent routing system for smart city applications using large-scale real-world traffic datasets. The project integrates data engineering, deep learning modeling, and algorithm optimization, targeting real-time prediction and route planning in dynamic urban environments.

Technical Highlights

- Processed and engineered **60k+ multivariate traffic records** including missing data imputation, temporal alignment, normalization, and spatial graph construction using Python (Pandas, NumPy).
- Developed a **hybrid LSTM–GCN** deep learning model capturing temporal trends and spatial road network structures, achieving an **8.6% accuracy increase**.
- Enhanced Dijkstra's algorithm with **dynamic time-weight adjustments** to support real-time routing under different traffic conditions.
- Evaluated model performance using RMSE, MAE, and MAPE, and optimized hyperparameters for better

generalization ability.

Key Achievements

- Achieved 8.6% improvement over traditional LSTM/GCN baselines.
- Reduced Dijkstra route computation time by 20% via dynamic weight optimization.
- Delivered a functional prototype capable of forecasting traffic and recommending optimized travel paths.
- Demonstrated strong capability in the design of machine learning, the analysis of mixed data in time and space, and the integrating of deep learning with classical algorithmic techniques.

Tech Stack

- Python, Excel, LSTM, GCN, Dijkstra's Algorithm, LaTeX

GitHub: github.com/Katherine-Tang/<CyberC-repo>

The Mathematical Contest in Modeling — Jan 2025

Algorithm Implementation & Programming Lead

Project Overview

Designed and implemented a predictive modeling system for Olympic medal distribution to forecast national performance and uncover key success factors. The project integrated hierarchical Bayesian modeling, time series analysis, and machine learning to analyze historical data, predict future outcomes, and provide strategic insights for Olympic committees.

Technical Highlights

- Developed a **Hierarchical Bayesian-ARMA** composite model with a Negative Binomial likelihood and Dirichlet distribution to predict total medals and gold-silver-bronze splits, incorporating host effect, national sporting ability, and temporal trends.
- Applied **K-Means++ clustering** on historical medal data to identify nations with high potential to win their first Olympic medal, and quantified their probability of success.
- Built **Multiple Linear Regression (MLR)** models to quantify the impact of event types, host advantage, and the "great coach" effect on medal counts, achieving R^2 values above 0.7.
- Utilized **Lasso regression** for feature selection to identify the most impactful sports for different countries, enhancing model interpretability.
- Employed **Simulated Annealing (SA)** with a dynamic penalty function to solve the coach allocation optimization problem, recommending optimal sports for coach investment in the USA, Japan, and Belarus.
- Authored the complete competition paper with **LaTeX**, including mathematical formulation, algorithm description, and visual presentation of results.

Key Achievements

- Accurately forecasted the 2028 Olympic medal table for top-competing nations using the Hierarchical Bayesian model, achieving a Mean Absolute Error (MAE) of 3.1592.
- Identified and quantified the breakthrough potential of 7 specific countries likely to win their first Olympic medal, providing actionable probabilities for targeted athletic investment.
- Delivered data-driven strategic insights by quantifying the host country advantage and the "great coach" effect, demonstrating its significant positive impact for some sports events.
- Effectively communicated complex methodology and strategic recommendations through a professionally typeset LaTeX report, providing clear, model-based guidance for national Olympic committees.

Tech Stack

- Python, Hierarchical Bayesian Model, ARMA, K-Means++, Lasso Regression, Simulated Annealing (SA), LaTeX

The 20th Citi Financial Innovation Application Competition — Dec 2024 – June 2025

Team Leader

Project Overview

Built a comprehensive Forex risk prediction system that integrates macroeconomic indicators, market data, and real-time financial news sentiment into a unified modeling framework. Designed the full end-to-end pipeline—from multi-source data ingestion and mixed-frequency alignment to machine learning modeling and real-time deployment—to support data-driven risk assessment and enhance decision-making in dynamic foreign exchange markets.

Technical Highlights

- Collected and merged multi-source financial datasets from **Wind**, **Bloomberg** APIs, and public economic databases; applied **MIDAS regression techniques** to align mixed-frequency (daily/monthly/quarterly) macroeconomic data.
- Built and optimized **XGBoost** and **Random Forest** classification models for Forex risk level prediction, achieving **75% precision** with feature importance analysis (volatility, inflation signals, capital flows).
- Developed a **news sentiment analysis pipeline** by scraping financial headlines, applying **FinBERT** for sentiment scoring, and integrating sentiment features to enhance model robustness.
- Built a production-ready **iOS mobile application** (SwiftUI) enabling real-time FX risk scoring with API-based model inference and live market data retrieval.
- Designed and managed **MySQL databases** for structured storage of macroeconomic datasets, model outputs, and user interaction logs; developed a responsive HTML dashboard for risk visualization and data querying.

Key Achievements

- Achieved **75% model precision** on Forex risk classification with strong generalization across multiple currency pairs.
- Delivered a fully functional **iOS real-time FX risk prediction app** by using Xcode, integrating ML inference and live data APIs.
- Improved end-to-end data processing efficiency by **50%** through database indexing, query optimization, and batch feature extraction.
- Enhanced predictive reliability and interpretability by incorporating sentiment signals, macroeconomic and market features.

Responsibilities Achievements

- Led a team of more than ten members, coordinated cross-functional efforts, and ensured project milestones delivery.
- Fostered a collaborative environment, encouraging creative problem solving and idea generation during weekly meetings.
- As the main contact person between the team and the organizing committee, ensure clear communication of project goals and progress.

Tech Stack

- Python, XGBoost, Random Forest, MIDAS Regression, FinBERT, HTML, MySQL, Wind Terminal, LaTeX
- iOS Development: SwiftUI, Xcode, RESTful APIs(ExchangeRate API, DeepSeek API)

GitHub: github.com/Katherine-Tang/<Citigroup-Cup-repo>

AWARD

IEEE CyberC Big Data Competition — October 2024
Second Prize (Top 3) — HKUST (Guangzhou)

The 20th Citi Financial Innovation Application Competition — June 2025
Second Prize (Top 10) — Citi China

The 20th Citi Financial Innovation Application Competition — June 2025
Special Prize (Top 1) — Citi China Markets

The Mathematical Contest in Modeling (MCM) — May 2025
Honorable Mention — COMAP

Contemporary Undergraduate Mathematical Contest in Modeling — November 2024
Jiangsu Province Second Prize — CSIAM

XJTLU Continuing Academic Merit Scholarship — July 2024

XJTLU Summer Undergraduate Research Fellowship, Student-nominated Winner — October 2025

2024 XJTLU Research-led Learning Symposium, Third Prize — November 2024

Microsoft Azure AI Fundamentals Certification May 2024

TECHNOLOGIES

Programming: Python (Pandas, NumPy, Scikit-learn), Java, C/C++, SQL, SwiftUI

Technologies & Tools: MySQL (schema design & indexing), Xcode, Wind Terminal, LaTeX, Cursor, Coze

ML & Data: XGBoost, Random Forest, FinBERT sentiment modeling, MIDAS regression, PyTorch

REFERENCE

Dr. Ren Zhao
Senior Teaching Fellow
School of Mathematics and Physics
Xi'an Jiaotong-Liverpool University
Email: Ren.Zhao@xjtlu.edu.cn
Phone: +86 512 8897 0257
Location: Suzhou, China

Dr. Soon Phei Tin
Teaching Fellow
School of Advanced Technology
Xi'an Jiaotong-Liverpool University
Email: Soon.Tin@xjtlu.edu.cn
Phone: +86 512 8188 9038
Location: Suzhou, China